

Assignment 4

Program -

```
1]
function areBracketsBalanced(expr)
{

    // Using ArrayDeque is faster
    // than using Stack class
    let stack = [];

    // Traversing the Expression
    for(let i = 0; i < expr.length; i++)
    {
        let x = expr[i];

        if (x == '(' || x == '[' || x == '{')
        {

            // Push the element in the stack
            stack.push(x);
            continue;
        }

        // If current character is not opening
        // bracket, then it must be closing.
        // So stack cannot be empty at this point.
        if (stack.length == 0)
            return false;

        let check;
        switch (x){
            case ')':
                check = stack.pop();
                if (check == '{' || check == '[')
                    return false;
                break;

            case '}':
                check = stack.pop();
                if (check == '(' || check == '[')
                    return false;
                break;

            case ']':
                check = stack.pop();
                if (check == '(' || check == '{')
                    return false;
                break;
        }
    }

    // Check Empty Stack
    return (stack.length == 0);
}

// Driver code
```

```
let expr = "([{}])";

// Function call
if (areBracketsBalanced(expr))
    document.write("Balanced");
    // .write("Balanced ");
else
    document.write("Not Balanced ");

// This code is contributed by rag2127

Output -

Balanced
```